

Weekly internship Reflection

University of Niagara Falls

Master's in data Analytics

Internship information

Intern name: Abigia Gashahun Edossa

Student Number: NF1021072

Company Information

Company name: Food Truck Association of Canada (FTAC)

Internship job title: Data Science & AI pipeline engineer Intern

Internship Supervisor: Mackenzie Peck

Week 2 reflection: 19th January - 23th January 2026

In the second week, I expanded my research into tokenization methods and classification strategies that could be applied to the dataset. I focused particularly on OpenAI tokenization, analyzing how it processes text into tokens and why it is a strong choice for this project. I learned how OpenAI tokenization aligns closely with GPT model behavior, making it effective for handling informal and conversational text such as Reddit posts. I also examined how tokenization can help estimate processing costs, filter overly long text entries, and improve consistency between preprocessing and model inference.

From my first meeting with my manager, we discussed possible technical approaches. We explored multiple strategies for structuring the classification process. One option involved using a ranking-based system to score how strongly a post expresses help-seeking intent, while another option focused on assigning a second category label to classify posts more clearly. We discussed the advantages and limitations of each approach, including accuracy, interpretability, and practical application within the project. This conversation helped guide the decision-making process regarding which classification method would be most effective.

Additionally, it was proposed to integrate OpenAI APIs into the project workflow to support tokenization, classification, ranking, and validation. We discussed potential API endpoints that could handle tasks such as detecting help-seeking content, scoring post relevance, and validating results using trained models. I also explored the idea of using a trained machine learning model as a validation layer to compare predictions against labelled data and ensure reliability.

Beyond the core project requirements, I began outlining optional research extensions, including sentiment analysis of help-seeking posts, intent detection to categorize business questions, topic modelling to identify recurring food truck discussion themes, and explainable AI techniques to better understand why certain posts are classified as help-seeking. These research directions were discussed with my manager as potential ways to enhance the project's depth and analytical value.

By the end of Week 2, I had developed a structured research plan, clarified technical methodologies with my manager, and gained a deeper understanding of how tokenization, classification, OpenAI APIs, and model validation could be integrated into a scalable and research-driven solution. This period significantly strengthened my ability to plan data-driven projects, communicate technical ideas clearly, and align research decisions with practical business objectives.

Student signature:

A handwritten signature in black ink, consisting of stylized, overlapping loops and strokes, likely representing the initials 'AG'.

Date: 24-01-2026